The verified reference model — GISS-E2-1-G

Status: computed and guarded — src/pacland/climate/reference_model.py + tests/test_climate_reference_model.py (9 tests, green).

The finding, in one sentence

> **Of 19 CMIP6 models with full conservation data, exactly one passes all three verification gates — GISS-E2-1-G — and its ECS, 2.71 K, is the single-model best prediction: the number from the one model we can fully verify.**

Why this model, and not the ensemble

The three gates are conservation (energy + mass + salt), sensitivity (ECS inside 2.5–4.0 K), and forcing (ERF2x inside 3.3–4.1 W/m²). Only GISS-E2-1-G — and its variant GISS-E2-1-G-CC — survives all three:

fact	value
axes passed	3/3
ECS	2.71 K
ERF2x	3.95 W/m²
conservation	0 ledgers leaked

The wider 15-model gate gives 3.08 K, and the IPCC best estimate is ~3.0 K. GISS's 2.71 K sits inside the same likely range, reached by the *strictest* possible gate — the single most-defensible number in the whole ensemble.

What "use the stable variant" means here

The model's prediction is its sensitivity (2.71 K per CO_2 doubling) — the physics it actually predicts. The 2100 warming is sensitivity **× the forcing pathway**, and the pathway is a *scenario input* (a label), not a model output. So the verified reference model pins the physics; the scenario is chosen by the question you ask.

Its own 2100 projection (Nazarenko et al. 2022)

The model's own SSP runs ([Nazarenko et al. 2022](https://doi.org/10.1029/2021MS002871), JAMES 14, e2021MS002871) give the verified model's answer to "how bad does it get", by 2100 relative to 1850–1880:

scenario	2100 warming
SSP5-8.5 (high emissions)	4.6–5.2 °C
SSP1-2.6 (strong mitigation)	below 2 °C (one variant 2.2 °C)
SSP1-1.9 (aggressive mitigation)	below 2 °C

Note the ECS method-dependence, stated rather than hidden: the model's own paper reports its effective sensitivity as 2.9–3.0 °C, while our gate uses Zelinka's cross-model-consistent 2.71 K. Both sit inside the IPCC likely range; the gap is estimation method (the Gregory regression vs the model's own

2×CO_2 run), not an error.

Honest limits

- **One family is one family.** GISS-E2-1-G's 2.71 K is a single model's value, not an independent consensus — its agreement with the IPCC best estimate is reassuring, not a proof.
- **The gate is strict by design.** It collapses 19 models to one because the other 18 fail at least one of conservation, sensitivity, or forcing. That is the point: the reference model is the one that does not fail.
- **This is a characterisation of the model, not a re-run.** No GISS simulation was performed; the verdict is assembled from Irving 2021 (drift) and Zelinka 2020 (ECS/ERF2x).

Reproduce

```
cd pacland
PYTHONPATH=src python3 -m pacland.climate.reference_model
PYTHONPATH=src python3 -m unittest tests.test_climate_reference_model -v   # 9 tests
```

Data: data/climate/cmip6_drift.json (Irving 2021) × data/climate/cmip6_ecs.json (Zelinka 2020). The three-axis verdict is src/pacland/climate/sensitivity_crosscheck.py.